

# AIZA — Malagasy/French Code-Switching Speech Benchmark

Sahara CodeSwitch Africa Challenge · Author: Arozo Andriamaharo (AIZA Project) · September 2026  
GitHub: <https://github.com/arozo-andria/aiza> · Web: [aiza.vercel.app](https://aiza.vercel.app)

## 1. EXECUTIVE SUMMARY & EVALUATED MODELS

Malagasy administrative interactions are characterized by pervasive intra-sentential code-switching: citizens embed French bureaucratic terms into Malagasy syntax (e.g. "Very ny CIN-ko, aiza no manao *déclaration de perte*?" ). This benchmark evaluates **Intron Sahara** alongside two open-source Malagasy fine-tunes on self-recorded authentic citizen queries in Antananarivo, Madagascar.

| Model                                                                | Architecture & Source                                                                    | Key Strengths                                                                                                  | Key Limitations                                                                               |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Intron Sahara<br>Shipped in AIZA                                     | Cloud API ( <code>infer.voice.intron.io</code> )<br>ASR mode: French ( <code>fr</code> ) | Fastest latency ( <b>3.2s avg</b> ); preserves French legal anchor keywords (" <i>déclaration de perte</i> "). | Malagasy unlisted in catalogue; treats Malagasy syntax as French phonetic homophones.         |
| Whisper-Small Malagasy<br>misterkissi/whisper-small-malagasy         | OpenAI Whisper-Small fine-tuned on Malagasy Common Voice (HF)                            | Accurate recognition of pure Malagasy grammar. Runs fully offline on CPU.                                      | High latency ( <b>6.5s avg</b> ); severely distorts French loanwords into Malagasy phonetics. |
| WAXAL Whisper Malagasy<br>waxal-benchmarking/whisper-small-waxal-mlg | Whisper-Small fine-tuned under African WAXAL Initiative (HF)                             | Best aggregate WER ( <b>76.4%</b> ); robust acoustic modeling for African accents. Offline.                    | Fails on code-switching transitions; requires custom tokenizer fallback.                      |

Ecosystem notes: **Rakitsoratra** (`rakitsoratra.professionnel.mg`) is Madagascar's commercial STT web app (MVola monetization) but closed-source. **Artifisialy / ABIDI v1** is a Malagasy Text-to-Speech (TTS) engine, not an ASR tool.

## 2. DATASET & METHODOLOGY

- Corpus:** 10 authentic administrative queries recorded via AIZA Studio ( `/record` ) covering lost IDs, birth certificates, marriage certificates, driver's licenses, address changes, passport fees, land titles, criminal records, and processing delays.
- Linguistic Structure:** 100% intra-sentential Malagasy/French code-switching.
- Audio Specifications:** 10 WebM/Opus recordings ( `01` to `10` ), 16kHz, total 62 seconds, 6.2s average length.
- Ground Truth:** Human-verified orthography normalized for Malagasy diacritics and French administrative terminology.

## 3. QUANTITATIVE RESULTS (LOWER WER/CER IS BETTER)

| Model                  | N  | Avg WER ↓ | Avg CER ↓ | Avg Latency ↓ | Downstream Procedure Match Rate ↑                   |
|------------------------|----|-----------|-----------|---------------|-----------------------------------------------------|
| Intron Sahara          | 10 | 82.6%     | 45.9%     | 3.2s          | 60% (6 / 10 queries matched to verified procedures) |
| Whisper-Small Malagasy | 10 | 88.2%     | 48.1%     | 6.5s          | 20% (2 / 10 queries matched)                        |
| WAXAL Whisper Malagasy | 10 | 76.4%     | 46.5%     | 6.8s          | 20% (2 / 10 queries matched)                        |

Downstream task note: Evaluates whether the transcript successfully triggers the correct verified administrative procedure card in AIZA's SQLite database ( `lost_cin` , `birth_certificate` , etc.).

4. UTTERANCE-BY-UTTERANCE BREAKDOWN (ALL 10 QUERIES)

| #  | Ground Truth (Reference)                                 | Intron Sahara (fr)                                                             | WAXAL Whisper Malagasy                                                                 | Whisper Malagasy                                                                             |
|----|----------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 01 | Very ny CIN-ko, aiza no manao déclaration de perte?      | 'Ferens Yenne même aux déclarations de perte'<br>WER: 80% · CER: 43%           | 'veriny se hianinaako aiza no manao dia kilao rasion deoparita'<br>WER: 80% · CER: 45% | 'verin'ny sehy ianinako, aiza ny manoha dia kilarasyon-deo mperita?'<br>WER: 100% · CER: 53% |
| 02 | Mila certificat de naissance aho ho an'ny zanako.        | 'La certificat de naissance Wançen à coût'<br>WER: 63% · CER: 38%              | 'lasary tia fikadonisao ho anny zanako'<br>WER: 75% · CER: 48%                         | 'na sertifika tena ison-sa ho an'ny zanako.'<br>WER: 63% · CER: 33%                          |
| 03 | Aiza no maka acte de mariage eto Antananarivo?           | 'Est-ce ma caracte de mariage étant en arrive'<br>WER: 88% · CER: 38%          | 'aisaka kay no maka haka tato mariazny eto antananarivo'<br>WER: 63% · CER: 38%        | 'haisakaainy makahakita teo maharihazy eto an-tananarivo?'<br>WER: 100% · CER: 42%           |
| 04 | Very ny permis de conduire-ko, inona no atao?            | 'Permet de conduire Jacques, inamo.'<br>WER: 78% · CER: 54%                    | 'tery mety ho gondohirako inona no atao'<br>WER: 67% · CER: 37%                        | 'Behery mety ho kanjo irako. Inano atao?'<br>WER: 89% · CER: 51%                             |
| 05 | Mila changement d'adresse aho, any amin'ny fokontany ve? | 'Le changement d'adresse unième fautin fait'<br>WER: 88% · CER: 50%            | 'lasandromandady iray sanga any aminny fokotany ve'<br>WER: 75% · CER: 44%             | 'oanjy monta adiraisa aho anemy fokitany ve'<br>WER: 75% · CER: 50%                          |
| 06 | Ohatrinona ny frais pour ny passeport vaovao?            | "Au chien n'en ferait pour pas sport fort?"<br>WER: 100% · CER: 50%            | 'ohatrany ny freimpasimbaovao vaovao'<br>WER: 71% · CER: 48%                           | 'ho tsy ny fray pora-pasiparva vao?'<br>WER: 100% · CER: 48%                                 |
| 07 | Tsy mety ny carte d'identité-ko, tapitra ny validité.    | 'Si méthan cachent d'identité accout obchant validité.'<br>WER: 89% · CER: 41% | 'tsy mety ny karita dido antitehako tapitra ny validite'<br>WER: 33% · CER: 18%        | 'tsy mety ny karita didon-titehako tapitra ny validy ite.'<br>WER: 56% · CER: 20%            |
| 08 | Manao ahoana ny procédure ho an'ny titre foncier?        | 'La procédure un titre français?'<br>WER: 75% · CER: 54%                       | 'posedure ho anny hita frantsay fotsy iaka e'<br>WER: 100% · CER: 77%                  | 'mprosesy hiraos anehititra firan-fonsiegle,'<br>WER: 100% · CER: 75%                        |
| 09 | Mila extrait de casier judiciaire aho, aiza no mankany?  | 'Mais il a extrait de gaz judiciaires Wacen Mcane?'<br>WER: 100% · CER: 45%    | 'milaiky sy reto ka azy avy ity sery aho aiza kay no mankany'<br>WER: 100% · CER: 57%  | 'milaheksitireto eo gazehajodisy herao aizakaian'ny makane?'<br>WER: 100% · CER: 60%         |
| 10 | Firy andro no ilaina hahazoana ny déclaration de perte?  | 'C'est rianoz à une déclaration de perte.'<br>WER: 67% · CER: 46%              | 'fy rihandro no azo ahoana dia kilao soa teo mpiata'<br>WER: 100% · CER: 54%           | 'firiandry noho aza hoana diaekilarasy ao teo perta?'<br>WER: 100% · CER: 48%                |

5. QUALITATIVE FINDINGS: THE CODE-SWITCHING ASYMMETRY PARADOX

**Key Discovery: Entity Preservation Beats Carrier Syntax in Downstream Tasks**  
While WAXAL Whisper achieved the lowest acoustic WER (76.4%), **Sahara achieved 3x higher downstream task matching (60% vs 20%)**. In administrative services, procedural intent is anchored in French terminology (*"déclaration de perte"*, *"certificat de naissance"*). Because Sahara recognized French phrases accurately, AIZA was able to reliably match procedures even when the Malagasy prefix was distorted into French homophones. In contrast, monolingual Malagasy models phonetically broke French words (e.g. *"déclaration de perte"* → *"kilarasyon-deo mperita"*), completely missing the keywords.

- Linguistic Failure Modes Observed:**
- **Sahara (French mode):** Map Malagasy syntax into French homophones (*"Very ny CIN-ko"* → *"Ferens Yenne même aux"*). Highly robust on French administrative compound nouns.
  - **Whisper Malagasy fine-tunes:** Accurately capture Malagasy grammatical particles (*"tsy mety ny"*, *"ho an'ny zanako"*), but lack vocabulary for French loanwords, creating phonetic hallucinations.

6. KEY ARCHITECTURE DECISIONS & ETHICS NOTE

- Technical & Architectural Choices**
- **Sahara as Primary Engine:** Selected for lowest latency (3.2s) and high legal keyword retention in live PWA.
  - **Mandatory Text Fallback:** Given the 82.6% WER on Malagasy, text typing is a first-class citizen in AIZA's UI.
  - **Deterministic Knowledge Grounding:** Extracted intents query an immutable SQLite database ( `data/aiza.db` ). AIZA strictly abstains (*"No verified procedure found"*) rather than fabricating legal advice.

- Ethics, Safety & Inclusion**
- **Informed Consent & Privacy:** All audio clips were self-recorded with informed consent. Zero audio is stored or retained on disk by the deployed PWA.
  - **Addressing Linguistic Inequity:** Malagasy (30M speakers) is absent from AfriSwitch and major speech benchmarks. Surfacing this real-world gap is an intentional contribution of this project.

7. REPRODUCIBILITY & OPEN SOURCE

The benchmark script, raw results, model configurations, and self-recorded audio clips are fully published and reproducible:

```
git clone https://github.com/arozo-andria/aiza.git && cd aiza
./benchmark/venv/bin/python3 benchmark/run_benchmark.py
```

Outputs: `benchmark/results.json` (raw metrics), `benchmark/REPORT.md` (full markdown report), `benchmark/audio/` (10 test clips).